

no trained network around it

a trained encoder or decoder around it

**the dynamics is fixed**

by design or at random

**Untrained, designed**

*nothing is fitted anywhere in the system*

Oscillator Ising machines  
(Hopf cochlea, the biological case)

**Untrained, with a trained readout**

*the readout is fitted, the dynamics is not*

ESN · LSM  
RON · Spintronic reservoir

**the dynamics is trained**

gradients flow through it

**Trained, designed**

*gradients reach the dynamics; at most a thin readout*

Un-0  
WONN

**Trained, inside a trained network**

*gradients reach the dynamics and its scaffolding*

AKOrN · KomplexNet · HoloGraph  
Kuramoto Attention · FSN · NWM  
coRNN · UnICORNN · LinOSS · D-LinOSS  
Kuramoto Orientation Diffusion